

Sudesh Subedi

Software Developer
Kathmandu, Nepal

www.sudeshsubedi.com.np
[mailto: faultypointer@proton.me](mailto:faultypointer@proton.me)
 [Github](#)

Education

TRIBHUWAN UNIVERSITY

IOE, Purwanchal Campus, Dharan, Nepal

Bachelor in Computer Engineering, 2022-2026

Major Project: Implementing Read/Write Overlay Filesystem Support in a Microkernel Based Operating System

Supervisor: Er. Manoj Kumar Guragai

Minor Project: A Federated Social Media for Scientific Communicators

Supervisor: Er. Pukar Karki

CAPITAL COLLEGE AND RESEARCH CENTER

Koteshwore, Kathmandu

National Examination Board +2, Science, 2019 - 2021

Projects

2025-2026

A Read/Write Overlay Filesystem Support in a Microkernel Based Operating System *Purwanchal Campus, Dharan, Nepal*

Worked on developing Overlay(Union) File System for Redox OS. It combines multiple different mount points of an existing filesystem into one, resulting in a single directory structure that contains underlying files and sub-directories from all sources.

2024-2025

A Federated Social Media for Scientific Communicators *Purwanchal Campus, Dharan, Nepal* 

Developed 3o14 (pi), a single user ActivityPub based fediverse client and server that supports rendering mathematical typesetting as a way to scientifically communicate between users, facilitating to write posts and replies using Markdown and LaTeX with a complete microblogging experience.

2024

A UCI Chess Engine 

Zig, Systems Programming

Developed a UCI-compatible chess engine in Zig with support for core UCI commands, move generation, board evaluation, and game state management. Focused on low-level systems programming, engine architecture, and search logic while exploring performance-oriented design in Zig.

Mini Projects

RustyJP 

A CLI tool for learning Japanese alphabets (Hiragana and Katakana) as well as Kanji using spaced repetition method with features for adding Kanjis yourself.

Zhip-8 

CHIP-8 emulator written in zig featuring opcode decoding, memory management, display rendering, timers, and input handling.

PngMe 

Rust-based CLI application for PNG steganography, supporting message encoding/decoding, chunk listing, and safe removal of embedded data within PNG file structure.